Surface salt bridges raise the thermal stability of an example lipase

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Fictional laboratory · prepared for the Hikari website; all measurements are synthetic

ABSTRACT — Lipases used at elevated temperatures lose activity as their lid domain unfolds. We introduced two charge pairs on the lid surface of an example lipase to form salt bridges with neighboring helices. Variant A melted 6.4 °C higher than wild type in a thermal shift assay (59.4 °C versus 53.0 °C) and retained full activity at 37 °C. After 10 min of heating, Variant A kept half of its activity up to 58.9 °C, compared with 52.6 °C for wild type. A single charge pair (Variant B) gave an intermediate effect. Surface salt bridges are a simple, activity-neutral route to more stable variants of this enzyme.

1. Introduction

Many industrial lipase reactions run above 50 °C, where unfolding of the lid domain limits enzyme lifetime. Stabilizing mutations often cost activity because they rigidify regions that move during catalysis. Salt bridges placed on the solvent-exposed face of the lid can add stability without touching the active site.

Here we tested whether two engineered charge pairs stabilize an example lipase, and whether the gain holds across three readouts: melting temperature, residual activity after heating, and activity at 37 °C.

2. Results

2.1 Thermal shift

Variant A melted at 59.4 ± 0.3 °C, 6.4 °C higher than wild type (53.0 ± 0.2 °C). Variant B, carrying a single charge pair, melted at 56.1 ± 0.4 °C (Fig. 1).

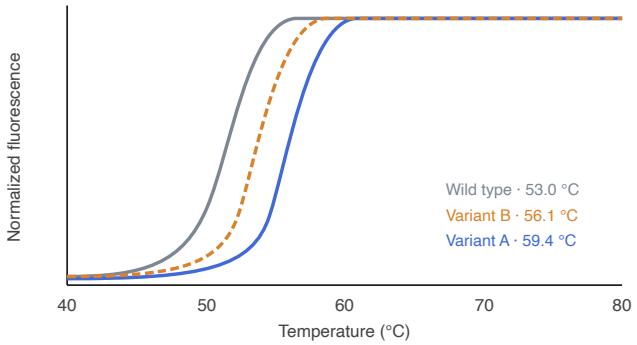


Fig. 1. Thermal shift melting curves (means of triplicate wells). Variant A shifts the melt by 6.4 °C.

2.2 Residual activity

After 10 min at each temperature, Variant A kept half of its activity up to 58.9 °C and wild type up to 52.6 °C (Table 1). At 37 °C, both enzymes had the same specific activity within error.

Enzyme	Tm (°C)	T50 (°C)	Activity at 37 °C
Wild type	53.0	52.6	100%
Variant A	59.4	58.9	98%
Variant B	56.1	55.8	101%

3. Discussion

Two surface salt bridges stabilized the lid without measurable cost to activity. The agreement between melting temperature and residual activity suggests the gain reflects slower unfolding rather than a change in dye binding.

The shift was measured on one preparation of each enzyme. An independent preparation is needed before choosing a lead variant.

4. Methods

4.1 Expression and purification

His6-tagged variants were expressed in an example *E. coli* strain, purified on Ni-NTA resin with a 20 mM imidazole wash and 250 mM imidazole elution, and desalted into assay buffer. Protein concentrations were matched before each assay.

4.2 Thermal shift assay

Each protein was mixed with a fluorescent protein dye and heated from 25 °C to 95 °C at 1 °C/min while fluorescence was recorded. Every variant was measured in triplicate wells, and Tm was taken at the maximum of the first derivative of the melt curve.

Three controls ran on every plate: wild-type enzyme as the reference melt; buffer with dye and no protein, to set the fluorescence baseline; and protein without dye, to rule out intrinsic fluorescence over the ramp.

4.3 Residual activity

Aliquots were held for 10 min at 40–73 °C in 3 °C steps, cooled on ice, and assayed with p-nitrophenyl butyrate at 37 °C. Activity is reported relative to an unheated aliquot. T50 is the temperature at which half of the activity remains, from a logistic fit.

4.4 Statistics

Values are means ± standard deviation of triplicate wells. Differences between variants were compared with a two-sided t-test.

References

1. Example Structural Biology Lab. Charge networks and unfolding kinetics in lipase variants. *Illustrative preprint* (2025).
2. Example Methods Group. Screening thermostable enzymes by differential scanning fluorimetry. *Illustrative preprint* (2024).

This document is fictional. It was written to demonstrate Hikari's paper tools and contains no real data.